

Range of all primitive datatype

- Integers** are typically stored with either 4 or 8 bytes
 4 bytes can store numbers between -2147483648 and 2147483647
 8 bytes can store numbers between -9223372036854775808 and 9223372036854775807
- Adding in binary is just like normal addition with carrying
 But when you run out of bits you can't carry anymore
 Leftmost bit indicates sign, so carrying to the leftmost bit changes a number from positive to negative.
 So adding 1 to 2147483647 goes to -2147483648!
 Called **Integer Overflow**
 - byte is the primitive type
 - Byte is the corresponding wrapper class
 - Byte.SIZE gives the number of bits in a byte
 - Byte.MIN_VALUE and Byte.MAX_VALUE give the minimum and maximum values a primitive byte can hold

```

1 public class DataTypesDemo {
2     public static void demonstrate() {
3         int a = 7;
4         int b = 5;
5         int intSum = a + b;
6
7         float x = 3.5f;
8         float y = 2.2f;
9         float floatSum = x + y;
10
11         System.out.println("Integer addition: " + a + " + " + b + " = " + intSum);
12         System.out.println("Float addition: " + x + " + " + y + " = " + floatSum);
13         System.out.println();
14         displayPrimitiveSizesAndRanges();
15     }
16
17     private static void displayPrimitiveSizesAndRanges() {
18         System.out.println(x: "Primitive type sizes and ranges:");
19         System.out.println("byte:   size = " + Byte.SIZE + " bits, range = " + Byte.MIN_VALUE + " to " + Byte.MAX_VALUE);
20         System.out.println("short:  size = " + Short.SIZE + " bits, range = " + Short.MIN_VALUE + " to " + Short.MAX_VALUE);
21         System.out.println("int:    size = " + Integer.SIZE + " bits, range = " + Integer.MIN_VALUE + " to " + Integer.MAX_VALUE);
22         System.out.println("long:   size = " + Long.SIZE + " bits, range = " + Long.MIN_VALUE + " to " + Long.MAX_VALUE);
23         System.out.println("float:  size = " + Float.SIZE + " bits, range = " + Float.MIN_VALUE + " to " + Float.MAX_VALUE);
24         System.out.println("double: size = " + Double.SIZE + " bits, range = " + Double.MIN_VALUE + " to " + Double.MAX_VALUE);
25         System.out.println("char:   size = " + Character.SIZE + " bits, range = " + (int) Character.MIN_VALUE + " to " + (int) Character.MAX_VALUE);
26         System.out.println(x: "boolean: size = 1 bit (logical), values = true or false");
27     }
28 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Float addition: 3.5 + 2.2 = 5.7

Primitive type sizes and ranges:

```

byte:   size = 8 bits, range = -128 to 127
short:  size = 16 bits, range = -32768 to 32767
int:    size = 32 bits, range = -2147483648 to 2147483647
long:   size = 64 bits, range = -9223372036854775808 to 9223372036854775807
float:  size = 32 bits, range = 1.4E-45 to 3.4028235E38
double: size = 64 bits, range = 4.9E-324 to 1.7976931348623157E308
char:   size = 16 bits, range = 0 to 65535
boolean: size = 1 bit (logical), values = true or false

```